Zepto Inventory Analysis

Objective

The objective of this project is to analyze product-level data from Zepto to understand pricing strategies, discount patterns, inventory distribution, and potential revenue across product categories using SQL.

Dataset Summary

- **Rows:** Total number of product records (SKUs) available in the dataset.
- **Columns:** 10 variables describing product details, pricing, discounts, inventory availability, and product specifications.
- **Key Features:**
    - SKU ID (unique product identifier)
    - Product category and name
    - Maximum Retail Price (MRP)
    - Discount percentage and discounted selling price
    - Available quantity in inventory
    - Product weight (in grams)
    - Stock availability status (out of stock)
    - Product quantity per unit

These variables were used to analyze **pricing strategies, discount patterns, product availability, and potential revenue across product categories**.

Data Cleaning

Removing Invalid Prices: Products with zero price were removed because they indicate incorrect data.

Converting Paise to Rupees: Prices were originally stored in paise and converted into rupees.

Data Analysis & Insights

1. Top 10 Best Value Products

	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dukes Waffy Orange Wafers	45.00	51.00
2	Dukes Waffy Chocolate Wafers	45.00	51.00
3	Dukes Waffy Strawberry Wafers	45.00	51.00
4	RRO Mozzarella Block Cheese	295.00	50.00
5	Ceres Foods Laal Maas Instant Liquid Mas...	220.00	50.00
6	Chef's Basket Durum Wheat Elbow Pasta	160.00	50.00
7	RRO Mozzarella Pizza Cheese	275.00	50.00
8	Ceres Foods Fish Mustard Instant Liquid M...	220.00	50.00
9	Dukes Waffy Strawberry Roll	150.00	50.00
10	Ceres Foods Nalli Nihari Instant Liquid Mas...	220.00	50.00

Insight

- These products offer the highest discounts and are likely to attract price-sensitive customers.
- Businesses can promote these items as featured deals or promotional products.

2. High MRP Products That Are Out of Stock

	name character varying (150)	mrp numeric (8,2)
1	Patanjali Cow's Ghee	565.00
2	MamyPoko Pants Standard Diapers, Extra Large (12 - 17 ...	399.00
3	Aashirvaad Atta With Mutigrains	315.00
4	Everest Kashmiri Lal Chilli Powder	310.00

Insight

- Several high-value products are currently out of stock.
- This indicates potential lost revenue opportunities, especially for premium products.

Recommendation:
Improve inventory planning for high-value items.

3. Estimated Revenue by Category

	category character varying (120)	total_revenue numeric
1	Fruits & Vegetables	10846.00
2	Meats, Fish & Eggs	20693.00
3	Biscuits	25007.60
4	Beverages	55051.00
5	Dairy, Bread & Batter	55051.00
6	Health & Hygiene	64180.00
7	Home & Cleaning	122661.00
8	Ice Cream & Desserts	224385.00
9	Chocolates & Candies	224385.00
10	Packaged Food	224385.00
11	Personal Care	270849.00
12	Paan Corner	270849.00
13	Munchies	337369.00
14	Cooking Essentials	337369.00

Insight

- This analysis estimates how much revenue could be generated from each product category based on available inventory.
- Categories with higher values represent strong revenue drivers.

4. High Price but Low Discount Products

Insight

- Some expensive products receive minimal discounts.
- These items may have lower price competitiveness, which can affect customer purchase decisions.

Recommendation:
Consider moderate discounting to improve sales.

5. Categories With Highest Average Discount

	category character varying (120)	avg_discount numeric
1	Fruits & Vegetables	15.46
2	Meats, Fish & Eggs	11.03
3	Ice Cream & Desserts	8.32
4	Chocolates & Candies	8.32
5	Packaged Food	8.32

Insight

- These categories rely heavily on discounting as a marketing strategy.
- High-discount categories may attract deal-seeking customers**.**

6.Best Value Products Based on Price per Gram

Insight

- This identifies products offering the lowest cost per gram, representing the best value purchases.
- Useful for customers comparing bulk value.

7. Product Size Segmentation

Products were grouped into weight categories.

Insight

- This segmentation helps analyze product packaging strategies and customer buying patterns.

8. Total Inventory Weight by Category

	category character varying (120)	total_weight bigint
1	Meats, Fish & Eggs	48016
2	Biscuits	84431
3	Fruits & Vegetables	91794
4	Health & Hygiene	142904
5	Dairy, Bread & Batter	143735
6	Beverages	143735
7	Paan Corner	348187
8	Personal Care	348187
9	Home & Cleaning	373161
10	Ice Cream & Desserts	490797
11	Chocolates & Candies	490797
12	Packaged Food	490797
13	Munchies	1404654
14	Cooking Essentials	1404654

Insight

- Shows which categories hold the largest inventory volume.
- Useful for warehouse management and supply chain planning